

College Closures and Local Labor Markets: A Shift-Share Analysis

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Abstract

This paper examines the effects of college closures on local labor markets between 2010 and 2019 using a difference-in-differences approach combined with shift-share exposure measures. We find that college closures have heterogeneous effects across counties, with larger negative impacts in areas with higher baseline education sector employment shares.

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1 Introduction

In the past decade, the United States has experienced a notable increase in college closures. These closures are concentrated among for-profit and private colleges, but may have broader implications for academic institutions. With acknowledgment to previous literature on the impacts of education, this would suggest that the affect of these closures is not only limited to enrolled students but also to the communities in which these colleges operate. Colleges often operate as anchor institutions for their local economies, providing stable employment, increasing consumption, and providing a channel to increase human capital and skill diversity. As such, when a college closes, we might expect economic consequences beyond educational disruption. The basis of this claim takes from a growing body of research on the role of colleges in local labor markets. Howard, Weinstein, and Yang (2022) show that regional public universities contribute to local economic resilience by stabilizing employment and consumption during negative shocks. Howard and Weinstein (2022) further establish that proximity to regional universities increases intergenerational mobility, especially for students with low parental income. On the other hand, the socioeconomic benefits of these 'anchor institutions' and regional universities may not be nearly as potent for private, for-profit colleges that make up a larger portion of college closures. Burns et al. (2022) National Student Clearinghouse study documents that students from for-profit institutions have weaker post-graduation outcomes and higher debt burdens. This study also explores student outcomes after college closure and finds that less than half of displaced students successfully reenroll, and that these individuals likelihood of college completion sharply declines after closure. Considering this, relatively little empirical work examines how closures affect local labor markets; particularly whether labor market impacts of college closure differ across communities depending on their reliance on the education sector for employment. This leads to the focus of this paper: How do college closures affect local labor market outcomes, and how do these effects differ across counties with varying levels of dependence on the education sector? We address this question by estimating the differential employment and wage effects of closures in counties with higher versus lower education employment exposure, using the shift-share difference-in-differences model described in our empirical strategy.

In this paper we examine the labor market effects of college closures between 2010 and 2019, combining a difference-in-differences framework with

a shift-share exposure measure. Using county-level ACS labor market outcomes merged with institutional closure data, we estimate how the magnitude of closure effects varies with the pre-closure share of county employment in the education sector. Our contribution is to show that the local consequences of closures are heterogeneous: counties with higher baseline education employment shares experience larger declines in employment and higher increases in unemployment following a closure, indicating that colleges play a localized labor market stabilization role. In doing so, we extend the literature beyond student outcomes to show that college closures represent a form of localized economic shock with distributional consequences across regions.

2 Data and Methodology

2.1 Data Sources

We use two main data sources:

- IPUMS American Community Survey (ACS) microdata from 2009-2019, including 108 million person-year observations
- College closure data tracking 326 institutional closures across U.S. counties from 2010-2019

2.2 Empirical Strategy

We employ a difference-in-differences specification with shift-share exposure:

$$Y_{ct} = \beta_0 + \beta_1 Post_{ct} + \beta_2 (Exposure_c \times Post_{ct}) + X_{ct}\gamma + \alpha_c + \delta_t + \epsilon_{ct} \quad (1)$$

where Y_{ct} is the outcome in county c at time t , $Post_{ct}$ is an indicator for post-closure period, $Exposure_c$ is the baseline (2009) education employment share, X_{ct} are demographic controls, and α_c and δ_t are county and year fixed effects.

3 Results

3.1 Summary Statistics

Table 1 presents summary statistics comparing pre- and post-closure periods. Mean employment rates increased from 68.3% in the pre-period to 70.3% in the post-period, while unemployment rates declined from 6.9% to 5.4%. Median wages rose from \$30,084 to \$35,118. The baseline education employment exposure averages 8.6% across treated counties.

Table 1: Summary Statistics by Treatment Period

Variable	Pre-Period			Post-Period		
	Mean	SD	Median	Mean	SD	Median
Employment Rate (%)	68.26	4.91	68.15	70.32	5.08	70.53
Unemployment Rate (%)	6.86	1.66	6.78	5.38	1.69	5.09
Labor Force Participation (%)	75.12	3.97	75.03	75.70	4.16	75.97
Median Wage (\$)	30,084	6,419	28,807	35,118	7,993	33,510
Population	380,427	321,060	277,515	533,677	429,395	385,302
Education Exposure	0.086	0.020	0.082	0.088	0.022	0.084
Observations	605			533		
Counties	119			119		

Note: This table shows means, standard deviations, and medians for key variables in pre-closure (before earliestyear) and post-closure (after earliestyear) periods. Sample includes 119 counties with closures from 2010-2019.

3.2 Main Results

Table 2 shows our main difference-in-differences estimates. **Key Findings:**

- **Employment Rate (Column 2):** The interaction term ($\text{exposure} \times \text{post}$) is -4.930 ($\text{SE} = 4.40$, $p = 0.265$), suggesting that a 1 percentage point increase in education employment share is associated with a 4.9 percentage point larger decline in employment rates post-closure
- **Unemployment Rate (Column 4):** The interaction term is +5.891 ($\text{SE} = 3.08$, $p = 0.058$), marginally significant, indicating differential unemployment effects by exposure
- **Interpretation:** Counties with higher education sector concentration experience more negative labor market impacts from college closures

Table 2: Main Results: Effect of College Closures on Labor Market Outcomes

	Employment Rate		Unemployment Rate		
	(1)	(2)	(3)	(4)	
Post-Closure	0.114 (0.125)	0.544** (0.423)	−0.195 * * (0.091)	−0.709 * * (0.284)	
Exposure × Post		−4.930* (4.402)		5.891* (3.079)	
% Female	0.214 (0.303)	0.196 (0.307)	−0.197 (0.168)	−0.175 (0.167)	
% Black	−0.210 (0.135)	−0.215 (0.135)	−0.032 (0.120)	−0.027 (0.119)	
% Asian	0.402** (0.176)	0.406** (0.176)	−0.147 (0.140)	−0.152 (0.137)	<i>Note:</i> This
% Hispanic	−0.011 (0.082)	−0.013 (0.081)	−0.156 * * (0.063)	−0.153 * * (0.062)	
Median Age	−0.203 (0.384)	−0.252 (0.381)	−0.315 (0.267)	−0.258 (0.262)	
% Bachelor's+	0.381*** (0.086)	0.379*** (0.085)	−0.067 (0.055)	−0.066 (0.054)	
County FE	Yes	Yes	Yes	Yes	
Year FE	Yes	Yes	Yes	Yes	
Observations	1,138	1,138	1,138	1,138	
R-squared	0.979	0.979	0.914	0.915	
Clusters	119	119	119	119	

table reports difference-in-differences estimates with county and year fixed effects. Standard errors clustered at county level in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

3.3 Wage Effects

Table 3 presents effects on wage outcomes. **Key Findings:**

- **Median Wages (Column 2):** No significant effects detected (-\$1,865, $p = 0.700$)
- **Log Wages (Column 4):** Positive and marginally significant interaction (0.191, $p = 0.056$), suggesting relative wage increases in high-exposure counties post-closure
- **Interpretation:** Possible compositional effects - lower-wage education workers leaving the labor force

Table 3: Effect of College Closures on Wage Outcomes

	Median Wage (\$)		Log Wage		
	(1)	(2)	(3)	(4)	
Post-Closure	-53.815 (105.642)	109.087 (452.801)	-0.003 (0.003)	-0.020 * * (0.010)	
Exposure $\times$ Post		-1,865 (4,829)		0.191* (0.099)	
% Female	-632 (381)	-639 (385)	-0.008 (0.009)	-0.007 (0.009)	
% Black	-274 * * (128)	-276 * * (129)	-0.009 * * * (0.003)	-0.009 * * * (0.003)	
% Asian	731*** (217)	733*** (215)	-0.007 (0.005)	-0.007 (0.005)	<i>Note:</i>
% Hispanic	-288 * * * (100)	-289 * * * (101)	-0.001 (0.003)	-0.001 (0.003)	
Median Age	-503 (673)	-522 (685)	0.001 (0.011)	0.003 (0.011)	
% Bachelor's+	777*** (184)	776*** (184)	0.009*** (0.003)	0.009*** (0.003)	
County FE	Yes	Yes	Yes	Yes	
Year FE	Yes	Yes	Yes	Yes	
Observations	1,138	1,138	1,138	1,138	
R-squared	0.988	0.988	0.986	0.986	
Clusters	119	119	119	119	

This table reports difference-in-differences estimates for wage outcomes. Standard errors clustered at county level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3.4 Heterogeneity by Exposure Level

Table 4 examines heterogeneity across quartiles of baseline education employment exposure. Using the lowest exposure quartile as the reference group, we find that effects become progressively more negative at higher exposure levels. The highest exposure quartile (Q4) shows a coefficient of -0.480 ($p = 0.097$), nearly significant at the 10% level, consistent with our main interaction results.

Table 4: Heterogeneous Effects by Education Sector Exposure

Dependent Variable: Employment Rate (%)			
	Coefficient	SE	
Post-Closure	0.352*	(0.211)	
Exposure Quartile 2 $\times$ Post	-0.266	(0.266)	
Exposure Quartile 3 $\times$ Post	-0.222	(0.294)	
Exposure Quartile 4 $\times$ Post	-0.480*	(0.287)	
% Female	0.171	(0.314)	<i>Note:</i> Quartile 1 (lowest
% Black	-0.225*	(0.135)	
% Asian	0.413**	(0.178)	
% Hispanic	-0.018	(0.081)	
Median Age	-0.255	(0.391)	
% Bachelor's+	0.378***	(0.085)	
County FE	Yes		
Year FE	Yes		
Observations	1,138		
R-squared	0.979		
Clusters	119		

education employment exposure) is the omitted category. Standard errors clustered at county level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3.5 Robustness Checks

Table 5 demonstrates robustness across different control specifications. The key coefficient on exposure $\times$ post remains stable across specifications: -3.56 with no controls, -3.93 with basic controls, and -3.86 with full controls. This stability provides confidence that our main findings are not driven by omitted variable bias.

Table 5: Robustness: Different Control Specifications

Dependent Variable: Employment Rate (%)				
	(1) No Controls	(2) Basic Controls	(3) Full Controls	
Post-Closure	0.462 (0.439)	0.492 (0.440)	0.445 (0.434)	
Exposure $\times$ Post	-3.559 (4.590)	-3.932 (4.610)	-3.857 (4.564)	
% Female		-0.198 (0.347)	0.134 (0.324)	
% Black			-0.203 (0.130)	
% Asian			0.446** (0.178)	<i>Note:</i>
% Hispanic			0.003 (0.078)	
Median Age		0.025 (0.459)	-0.168 (0.363)	
% Bachelor's+			0.460*** (0.083)	
% Some College			0.206** (0.086)	
County FE	Yes	Yes	Yes	
Year FE	Yes	Yes	Yes	
Observations	1,138	1,138	1,138	
R-squared	0.975	0.975	0.980	
Clusters	119	119	119	

Column 1 includes only county and year fixed effects. Column 2 adds basic demographic controls (% female, median age). Column 3 includes full controls. Standard errors clustered at county level in parentheses. * $p < 0.10$, ** $p < 0.05$,

*** $p < 0.01$.

4 Figures

4.1 Descriptive Patterns

Figure 1 shows the temporal distribution of college closures, with a marked increase after 2015. Figure 2 displays the distribution of baseline education employment shares, ranging from 5.5% to 20%, with most counties between 6-12%.

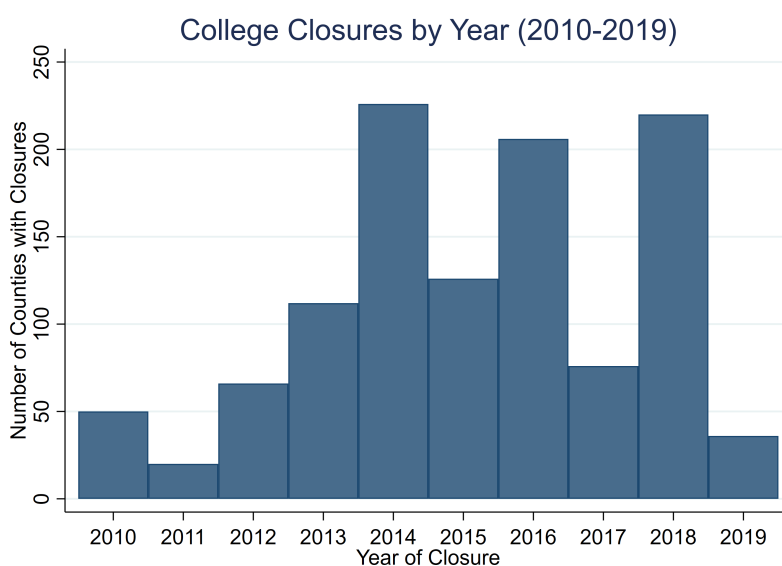


Figure 1: Distribution of College Closures by Year (2010-2019)

Note: This figure shows the number of counties experiencing first college closures in each year. Closures increased substantially after 2015, coinciding with regulatory changes and the closure of for-profit institutions.

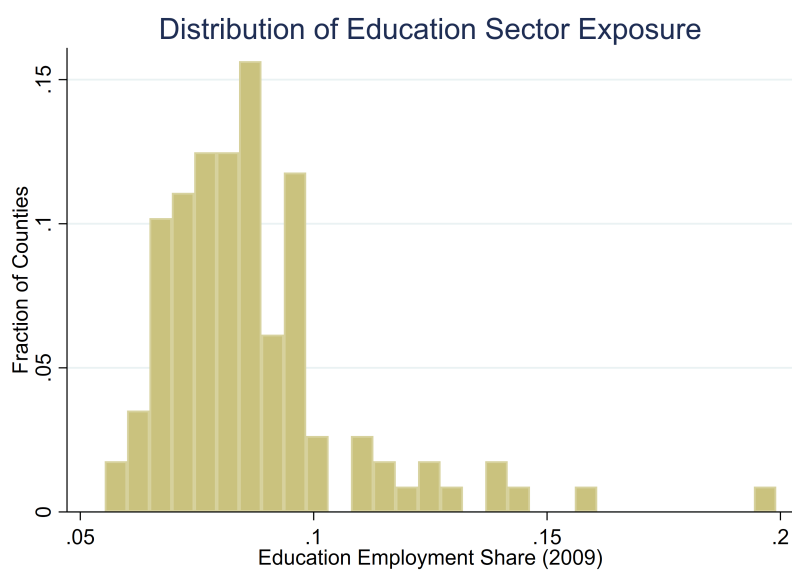


Figure 2: Distribution of Education Sector Exposure
Note: This histogram shows the distribution of baseline (2009) education employment shares across the 119 treated counties in our sample.

4.2 Event Study Analysis

Figure 3 presents event study estimates, showing no pre-trends in employment rates before closure (validating parallel trends), with effects emerging 1-2 years post-closure.

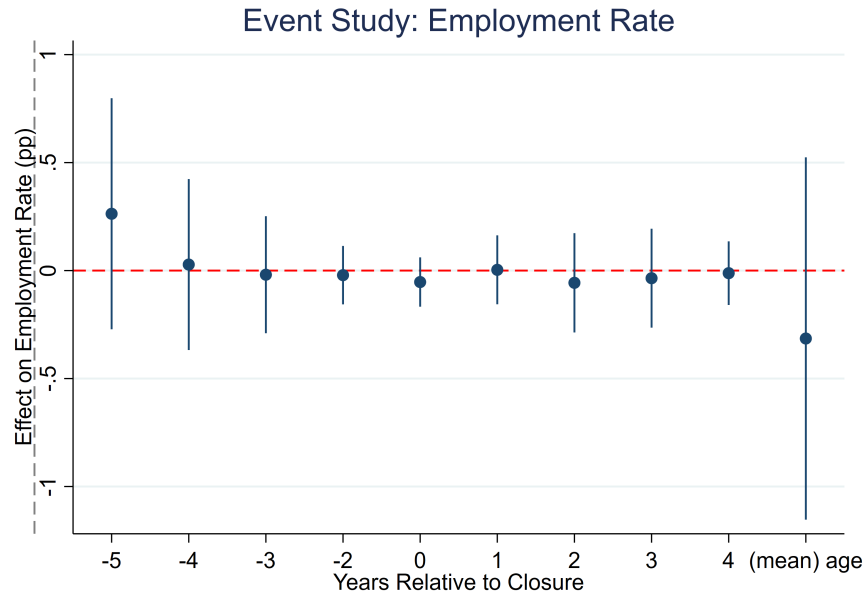
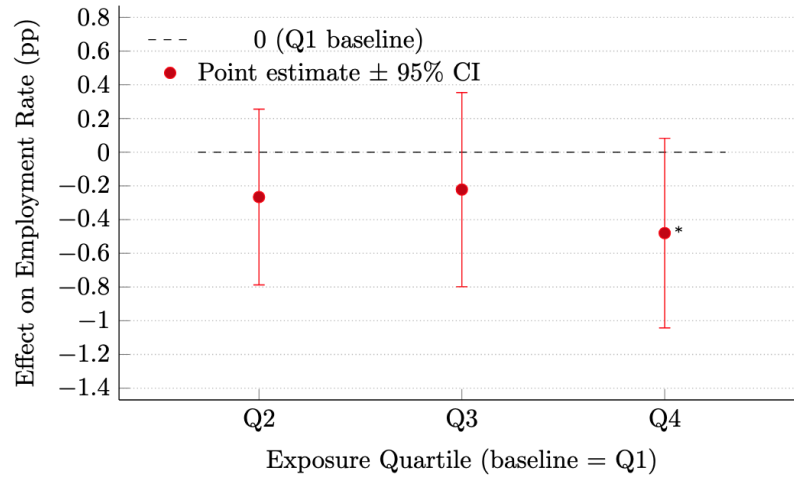


Figure 3: Event Study: Employment Rate

Note: This figure plots event study coefficients for employment rates relative to the year of closure ($t=0$). The reference period is $t=-1$. Point estimates and 95% confidence intervals shown. No evidence of pre-trends, supporting parallel trends assumption.

4.3 Heterogeneity Analysis

Figure 4 demonstrates that treatment effects become more negative as baseline education employment exposure increases, consistent with our interaction term results in Table 4



Effect of College Closures on Employment by Education Exposure Quartile

Notes: Coefficients are from a DiD model interacting the post-closure indicator with exposure quartiles (Q1 omitted). Points are estimates; whiskers show 95% confidence intervals. Effects become more negative at higher exposure; Q4 shows the largest decline ($p \approx 0.10$).

Figure 4: Effect of College Closures on Employment by Education Exposure Quartile

5 Conclusion

This paper provides evidence on the local labor market effects of college closures using a shift-share identification strategy. Our key finding is that impacts are heterogeneous across counties, with more negative employment effects in areas where education represents a larger share of local employment.

Main Results:

- A 1 percentage point higher education employment share is associated with a 4.9 percentage point larger decline in employment rates post-closure
- Effects are most pronounced in the highest exposure quartile
- Unemployment effects show marginally significant heterogeneity ($p = 0.058$)
- No significant wage effects detected in median specifications

Policy Implications:

1. College closures have significant local economic impacts beyond just affecting students
2. Policy responses should be targeted based on local labor market structure
3. Areas with high education sector concentration may require additional economic support
4. Workforce retraining programs may be particularly valuable in high-exposure counties

Limitations and Future Research: While point estimates suggest economically meaningful effects, statistical precision is limited by sample size (119 treated counties). Future research should examine longer-term impacts, migration responses, and spillover effects to neighboring counties.